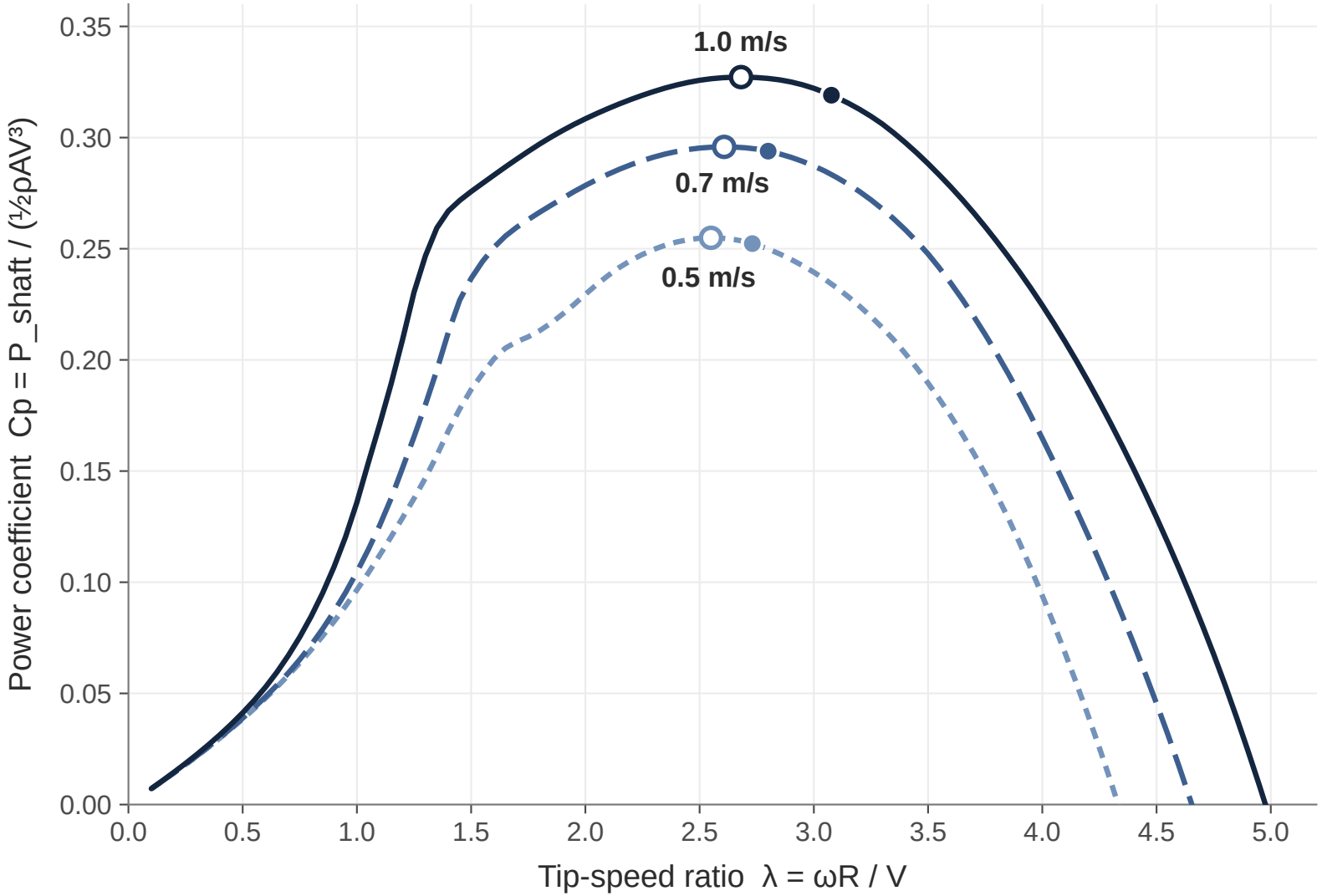
Current v5 rotor: power coefficient vs tip-speed ratio

3 blades, NACA 4412, 203.2 mm diameter, 28 mm hub · BEM model with NeuralFoil polars



Model predictions; experimental validation pending

Predicted values

Water speed	Peak Cp	λ at peak	λ operating
0.5 m/s	**0.25**	2.5	2.7
0.7 m/s	**0.30**	2.6	2.8
1.0 m/s	**0.33**	2.7	3.1

○ Peak Cp (maximum rotor efficiency)

● Operating point (GL60 + MPPT model)

Peak Cp rises with speed: the blade Reynolds number rises, so section drag falls.

The operating point sits slightly above the peak λ: MPPT maximises electrical output, and lower torque means lower winding loss.

Cp is referred to the full 203.2 mm disc area.